

BG4104 Hypothesis Assignment

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I. QUESTION 1

Tumour segmentation supports diagnosis, surgery, and radiotherapy, but the per-voxel tumour masks it requires are slow and expensive to make, taking up to 150 minutes per lesion [2], and are not part of a radiologist's everyday job. As a result, most public datasets have fewer than 100 tumour masks, and there is none that exceeds 2,000 per tumour type [1]. In contrast making radiology reports is in a radiologist's daily work, which exist in huge quantities (public datasets already exceed 75,000) [1]. Reports carry useful information such as tumour count, location, and size, but only at a classification and detection level, not the per-voxel tumour masks that segmentation needs. The jump from classification and detection (radiology report) to segmentation (per-voxel tumour mask) is therefore the gap the paper solves, by introducing R-Super, a training paradigm whose two new loss functions, volume and ball loss, convert text from a radiologist's report into per-voxel tumour masks. This is worth pursuing because AI involvement in segmentation has been shown to cut interaction time by around 33% while keeping comparable accuracy [3].

The paper also notes and solves smaller gaps left by earlier models as seen in Table 1 [1]. Standard segmentation and Models Genesis ignore reports, multi-task learning and CLIP-like methods read reports but only for auxiliary tasks whilst report-guided pseudo-labels reduce false positives but miss false negatives and ignore tumour size. R-Super is the only method that uses detailed report information from the radiologist's report to supervise segmentation directly and punish both error types, raising F1-Score by up to 16% over masks alone [1].

II. QUESTION 2

The authors make their hypothesis clear early in the introduction, stating that they "hypothesize that radiology reports can be used to directly supervise tumour segmentation." In simpler terms, they claim that details already written in a report, such as how many tumours there are and where and how large they are, can be converted into voxel-wise supervision and used to train the segmentation model (MedFormer). The expectation is that learning from reports this way will produce better segmentation than relying on the few available masks alone. What makes this specific, rather than a vague hope that reports might help, is the word "directly." The earlier methods in Table 1 used reports only for side tasks like classification or contrastive pre-training, so any benefit to segmentation was indirect. R-Super instead bets that a report can supervise the

segmentation output itself, which is exactly what the Volume Loss and Ball Loss are built to test.

Shown by the experiments, adding reports raised tumour detection F1-Score by up to 16% over mask-only segmentation, and R-Super outperformed five methods on both an internal test set (UCSF-Test) and an unseen external hospital (JHH-Test) [1]. Because the improvements held across few, medium, and many masks, it's fair to say that reports can supervise segmentation directly and effectively, just as the authors claimed.

III. QUESTION 3

I came into this paper with no real background in machine learning, deep learning, or medical imaging, so much of the vocabulary and many of the concepts were new to me. Understanding the authors' claims meant teaching myself the underlying ideas first, which I did mostly through Google searches and Claude, using them to explain unfamiliar terms and to re-explain the harder ideas until they made sense.

Firstly, I had to look up what tumour segmentation is and how it differs from classification and detection. Medical image segmentation is the process of extracting a region of interest, such as a tumour, from 3D scan data like a CT, producing a labelled mask of that region [4]. This gave me the context for the paper's motivation, since the gap only makes sense once you realise that segmentation means labelling the scan voxel by voxel, which is far more work than simply noting that a tumour is present.

Secondly, the harder material was the notation and the method itself. I needed to understand what a loss function is (the score for how wrong the model's prediction is, which it then learns to reduce), along with the two traditional losses the paper relies on, cross-entropy and the Dice Similarity Coefficient. The two new ideas, the Volume Loss and the Ball Loss, took longer, and the Ball Convolution in particular only made sense once I worked through small examples of how a fixed spherical kernel slides across the model's output to locate a tumour. To stop the notation feeling abstract, I took the paper's diameter-to-volume equation,

$$V = \frac{\pi}{6} \cdot d_1 \cdot d_2 \cdot d_3 \quad (1)$$

[1], and applied it to some made-up tumour diameters of my own. Since reports give diameters rather than volumes, working a volume out by hand was what made it click, as I could finally see how a single line of text in a report becomes a concrete target the model is trained against.

A further point that required clarification was the difference between R-Super and MedFormer, which I had initially assumed were the same thing. After looking into it, I understood that MedFormer is a pre-existing model that produces the voxel-wise output, while R-Super is the paper’s own contribution, a training method built from the two new losses that allows that model to learn from reports. This distinction mattered, since it is the difference between assuming the paper built a new model and recognising that it built a new way to train an existing one.

Finally, I had to understand the metrics before I could judge the claims. F1-Score was new to me, and learning that it combines precision and recall into one value gave the “+16%” improvement real meaning, since tumour detection has to balance catching real tumours against raising false alarms. I also checked the cost figures behind the motivation: fully manual segmentation can take from a few minutes up to 150 minutes per single lesion [2], and AI involvement in segmentation has been shown to cut interaction time by around 33% while keeping comparable accuracy [3]. These sources confirmed that the problem is real and worth solving, and looking all of this up was the main way I came to understand the paper.

IV. QUESTION 4

One limitation from this paper is that R-Super depends entirely on the information contained in radiology reports, yet these reports aren’t always consistent. Reporting styles, phrasing and way of writing differ a lot between radiologists, ranging from detailed free-text descriptions to short, question-oriented notes, and there is a well-documented lack of standardisation across the field [5]. A survey of referring physicians found wide variation in report length, structure, and completeness, with many reports missing specific details entirely [5]. This matters for R-Super because its losses rely on the report stating a tumour’s count, location, and size. If a radiologist phrases these differently, is missing a measurement, or follows a different convention, the supervision the model receives becomes weaker or inconsistent. The paper itself acknowledges this issue, noting that it did not apply its losses to organs where reports mentioned tumours but not their sizes, which affected under 11% of cases [1]. While this limitation can be partly reduced by training on a much larger amount of reports, so that the model averages over unique and inconsistent differences in a radiologist’s report, inconsistencies such as missing sizes cannot be fully resolved by the amount of training data of reports used.

Secondly, although R-Super is designed to reduce the field’s reliance on segmentation masks, it does not remove that reliance entirely. The method still has to be trained on some CT-Mask pairs to anchor the segmenter before the report-based losses are applied, using 50, 344, or 1,700 masks in the paper’s experiments [1]. The results also show that the benefit of adding reports was largest when masks were scarce or moderate and smallest when masks were already abundant (around +16% at 344 masks but only +4.3% at 1,700) [1]. This

suggests that R-Super narrows the mask restriction rather than fully solving it, and that performance is still tied to the amount and quality of mask data available. A larger and more diverse pool of CT-Mask scans would likely improve accuracy further, so the method’s ceiling is still partly set by the very scarcity it is trying to address.

Thirdly, a more discrete weakness is how the Ball Loss handles multiple tumours in the same organ. It locates them one by one, from largest to smallest, marking each tumour’s voxels and then removing them from the output so the next tumour is forced to a different region [1]. Because of this order, a single mistake can affect the rest. If the model gets the largest tumour’s location wrong, it removes the wrong voxels, and every smaller tumour after it is then matched against an output that has already been compromised. One early error doesn’t stay contained, it carries through to all the remaining tumours in that organ, and it does the most damage to the small tumours, which are already the hardest ones to detect in the first place.

Finally, like most deep learning methods, R-Super offers no explanation for its predictions. The model outputs a tumour location from millions of learned weights but cannot justify why it placed a tumour in a given spot [1]. In a clinical setting, where a radiologist may want a reason before trusting an AI’s output, the lack of interpretability is a real drawback.

V. QUESTION 5

One next step follows directly from the paper’s central claim. The authors argue R-Super can scale segmentation using the vast number of reports in hospitals, yet their experiments used only a few thousand tumour reports (around 5,000, from a UCSF set of roughly 6,700 CT-report pairs), while noting a single hospital can hold “orders of magnitude more” [1]. The scalability is therefore claimed but not demonstrated at the scale that matters. A clear next step is to train R-Super on hundreds of thousands of real reports to see whether the gains keep growing or see it plateau at a certain number of reports. This leads to my first hypothesis that R-Super’s accuracy gains continue to rise as training reports grow far beyond the few thousand used here, rather than levelling off early. It is testable by training the model on increasing report counts (e.g. 10K, 50K, 100K) and tracking the F1-Score. If it keeps rising, it supports the claim that reports are a scalable path to better segmentation; if it plateaus, the benefit of reports is more limited than implied, and mask quality remains the deciding factor.

A second step that I’m expanding on from question 4 is that, like most deep learning models, R-Super gives no explanation for why it predicts a tumour in a given location, which is an obstacle in a clinical setting where a radiologist may want a reason before doing something that affects a human life. Pairing R-Super with an interpretability method, such as a heatmap showing which regions most influenced each prediction, would help. This raises my second hypothesis that adding an interpretability layer would make predictions more

verifiable to clinicians while causing little or no drop in accuracy. This matters because in a health setting, undetected errors are especially dangerous; a confident but wrong prediction may be acted on by a clinician, and if such errors go unnoticed they can quietly persist or even be reinforced as the model continues to train on its own outputs. It could be tested by comparing two versions of the model, with and without the interpretability component, on both F1-Score and how often a radiologist agrees with the highlighted regions. If accuracy holds, interpretability and performance can coexist; if it falls, there is a real cost to making the model explain itself, which matters before clinical deployment.

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